# Critical Ping

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| **Team Owner** | Proc and Roll |
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- 2024-05-07 EM-64860: Critical Ping failure after SL1 upgrade - Drop File requested.


## Diagnosis and Mitigation

### Critical Ping configuration:

Critical ping uses Fping utility to ping multiple IP at a time. There are multiple configuration parameters which can affect the performance of the Fping/Critical Ping.

Following parameters are used with default values:

- Critical Ping Frequency : Disabled. User can select from 5 second to 2 min. poll frequency to enable it from Device Properties.
- Batch : 100 IP per fping call.
- Timeout : 1000 ms
- Packet Size: 56 kb
- Percentage : 100%  All pings to the IP should be successful.
- Ping_count: 1 ping to the IP.

There are global-system level threshold settings which can be accessed at : System > settings > Thresholds > ICMP Availability Thresholds.

System level settings you can also access directly on DB.
Ex: "Select avail_ping_count, avail_ping_req_pct, ping_packet_size from master.system_settings_thresholds

Device level settings can be accessed at : Device properties > Thresholds > ICMP Availability Thresholds

Device level settings you can also access directly on DB.

Ex: "Select critical_avail_ping_count, critical_avail_ping_req_pct, critical_ping_packet_size from master.system_settings_thresholds"

To use device level threshold need to use **critical_ping_share_avail_thresh** flag by manually updating it.
**More information about the thresholds settings is here :** Critical Ping Thresholds

**Troubleshooting:**

**Issues:** Mostly tickets we get about Critical ping is creating flip-flop events for devices and Performance issue. These are in general points to troubleshoot critical ping issue.

First thing we should analyze the above mentioned critical ping parameters/thresholds set for the affected devices. Fine tuning the critical ping configuration can resolve the issue.

- If only a set of devices has issue, use device level thresholds settings for the affected devices.
- Higher frequency of ping, higher no. of ping counts and high no. of devices aligned to ping at same frequency will affect performance of critical ping service.
- Avoid high values settings at system level threshold. Use device level settings for important devices. Let say if devices are aligned to 30 second critical ping frequency, the devices will be pinged at 30, 60, 90 and 120 second frequency poll. i.e. 2 pings in a minute. Change frequency to 60 second so it will ping the device only once in a minute and reduce load.
- Reduce no. of ping count if it is higher than 1,it will improve performance.
- If network has high latency, try increasing the timeout period from 1000ms to 2000ms.
- Also try reducing batch size less than 100. Start with 1 batch size if that shows improvement in the performance, adjust the value by some trial and error.
- Test affected device IP by fping from the collector CLI :

  **CLI**

```
/usr/sbin/fping -e -b56 -C 4 -t1000 -q 10.2.4.192 10.2.4.194
```

See if the devices are responding to fping.

- In logs you can check actual gap between two polls using following command:

**Logs**

```
[em7admin@mn tmp]$ grep -e "60 second critical polls" -e "processing 120 second critical polls" silo.log
```

- To check if there is problem in SL1 critical ping service, you can write a cronjob and execute fping with  same parameters as configured in SL1 and execute it for few min. on the collector . check the logs for performance and compare with SL1 critical ping results.

**Cronjob**

```
echo “processing 215 IP critical poll $(date)”  >> /var/log/em7/pingtest.log ; /usr/sbin/fping -e -b56 -C 1 -t1000 -q
10.2.8.88 10.2.6.32  2>> /var/log/em7/pingtest.log
sleep 30
echo “processing 215 IP critical poll $(date)”  >> /var/log/em7/pingtest.log ; /usr/sbin/fping -e -b56 -C 1 -t1000 -q
10.2.8.88 10.2.6.32  2>> /var/log/em7/pingtest.log
```

## Post Mortem Articles

### 2024-05-07 EM-64860: Critical Ping failure after SL1 upgrade - Drop File requested.

ts_guide_to_jira_link

Link: EM-64860: Critical Ping failure after SL1 upgrade - Drop File requested. CLOSED

From the description:

```
Our Platform was upgraded yesterday and we now have a huge amount of critical errors which are reported in the system logs.  we were sent a link to a known bug and we would request a drop file to fix this as we have
over 80 collectors to change if we have to enter the workaround on each collector.   The link we were sent was https://support.sciencelogic.com/s/article/13079 and the error we are seeing in out logs are as below .

silo.snmp_detail_collect.24: Unhandled exception on appliance: 266, process Data Collection: SNMP Detail:
Traceback (most recent call last):
File &quot;/tmp/pip-install-PbZ5ZM/sl-process/sl_process/proc_pda.py&quot;, line 214, in _init_
```

```
File &quot;/tmp/pip-install-PbZ5ZM/sl-sysuptime/sl_sysuptime/collect_snmp_detail.py&quot;, line 155, in main
File &quot;/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py&quot;, line 134, in run_collection
File &quot;/tmp/pip-install-PbZ5ZM/sl-collect/sl_collect/shared.py&quot;, line 146, in _run_collect_sync
File &quot;/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py&quot;, line 47, in new_f
ChildProcessException: Traceback (most recent call last):
File &quot;/tmp/pip-install-PbZ5ZM/sl-process/sl_process/exceptions.py&quot;, line 43, in new_f
File &quot;/tmp/pip-install-PbZ5ZM/sl-sysuptime/sl_sysuptime/collect_snmp_detail.py&quot;, line 86, in collect
File &quot;/tmp/pip-install-PbZ5ZM/sl-credentials/sl_credentials/snmp.py&quot;, line 79, in snmph_from_cred_array
File &quot;/opt/em7/lib/python/future/types/newstr.py&quot;, line 102, in _new_
return super(newstr, cls)._new_(cls, value)

Reason for Escalation: Engineering assistance needed to diagnose issue in order to find workaround
Which troubleshooting doc(s) were reviewed: https://support.sciencelogic.com/s/article/13079
```

In this comment EM-64860: Critical Ping failure after SL1 upgrade - Drop File requested. CLOSED

```
I believe the title of this Jira is a bit misleading.
I talked to @Carlos Rojas Plata about this & we believe this is another case of the following known issue: EM-64236
Since the Proc N Roll team fixed that ticket I am going to see about forwarding this escalation to them so they can assist further.
```

That support defect is referenced in this TSG section: Credentials Troubleshooting | 2024 04 09 EM 64236: UnicodeDecodeError: 'ascii'-codec-can't-decode-byte-0xf1-in-position-1:-ordinal-not-in-range(128)